

AI Structuralism

Representation has generally been understood as a process of making the world visible through images, symbols, and systems of depiction. However, from Renaissance perspective to contemporary computational imaging, visual media have not simply sought to capture reality with increasing precision, but also have repeatedly exposed the idea of representation as a constructed framework through which reality is organized, translated, and often contested. Across mimetic, abstract, and computational practices, representation emerges less as a transparent mirror of the world than as a constructed system shaped by perception, structure, and interpretation. As Immanuel Kant proposes, experience is mediated through a priori structures that shape how the world can appear to us (Kant, 1998). What we perceive as visual truth is often the result of interpretive structures rather than direct correspondence with the world.

Collapse, as an ontological condition of the visual image and the disintegration of representational form, can be understood as the process through which an image begins to lose its recognizable appearance, revealing that meaning is not fully contained in visual surface but emerges through underlying structures and interpretive processes. This project is specifically concerned with the ontology of visual representation: how images produce meaning, how recognition persists through abstraction, and how perception operates when visual resemblance begins to break down. From a structuralist perspective, collapse does not necessarily imply the disappearance of meaning; rather, it exposes the relational systems that organize perception and signification. Structuralism proposes that meaning is produced through differences and relationships within structured systems of signs rather than through direct correspondence with reality (Saussure 1959). In this sense, when representation collapses—when detail, resemblance, or descriptive completeness diminishes—the structural relations that sustain recognition may remain operative. However, post-structuralist thought complicates this view by arguing that such structures are neither stable nor universally determining. Post-structuralist

thought challenges the stability of representation by proposing that meaning is not fixed, but continually produced and transformed through context, interpretation, and historical conditions (Derrida 1976). The collapse of visual resemblance and the assumption that an image can stably convey meaning therefore does not simply reveal an underlying structural truth, but also exposes the instability of meaning itself, suggesting that visual representation operates within a dynamic field in which structure and interpretation continuously interact.

In the current era of generative artificial intelligence, the structuralist idea that meaning emerges from relational systems rather than direct correspondence with reality takes on renewed relevance. Generative AI models such as diffusion networks and large language models do not perceive or understand the world through embodied experience; instead, they learn statistical relationships between elements across vast datasets of images and texts. What these systems produce are not representations grounded in direct observation, but configurations assembled from patterns of difference, similarity, and association (Underwood 2019). In this sense, AI operates through a fundamentally structural logic: objects are encoded as distributions of features, semantic categories are formed through relational proximity within latent space, and meaning is inferred from correlations rather than intrinsic properties. However, the outputs of generative AI also challenge classical structuralism by demonstrating how these relational structures are inherently unstable, probabilistic, and shaped by the biases of training data. The image generated by AI may appear coherent and realistic, yet it is ultimately a synthetic construction that refers not to the world itself but to an accumulated structure of prior representations (Paglen, 2016). As a result, generative AI makes visible the degree to which contemporary representation is mediated through abstract systems of relations, foregrounding both the explanatory power and the limitations of structuralist accounts of meaning.

To examine structuralism within the context of contemporary AI, I developed a series of studies including a reconstructed 3D model of a lemon, a time-lapse projection visualizing the

transition from latent diffusion noise to image formation, and a real-time demonstration of AI misrecognition.

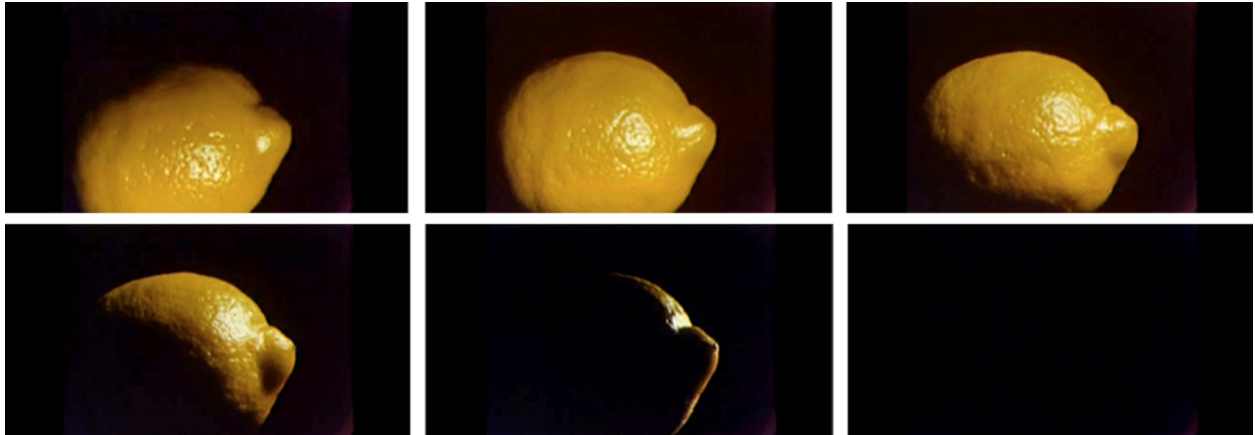


Figure 1. Still from *Lemon*, directed by Hollis Frampton, 1969.

The lemon experiment originates from the structural film *Lemon* (fig.1, 1969) by Hollis Frampton, which presents a single lemon illuminated from varying angles over the course of seven minutes. The film proposes that what viewers perceive as a coherent object is not the lemon itself but an accumulation of changing light conditions, color gradients, and perceptual continuity. In other words, the identity of the object emerges from relational visual information rather than from direct material presence.

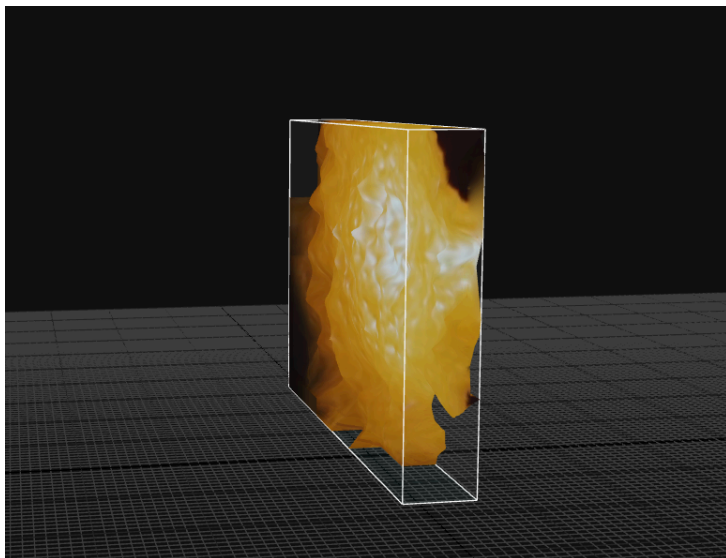


Figure 2. 3D photogrammetric reconstruction of the lemon generated from frames extracted from Lemon

Using frames extracted from the film, I generated a 3D reconstruction through photogrammetry (fig2.). The resulting model does not resemble a volumetric citrus fruit but instead appears as a flattened disk with a jagged, spiked topology. This discrepancy reveals a fundamental divergence between human perception and machine perception. Human viewers recognize the lemon not only through visual cues but also through prior knowledge, embodied experience, and cultural associations—the expectation of weight, texture, smell, and taste. The machine, by contrast, does not possess such experiential frameworks; it constructs the object solely through mathematical correspondences between image features. Photogrammetry operates through processes such as feature detection, image alignment, triangulation, and point cloud generation, identifying shared coordinates across multiple images in order to infer spatial structure. The resulting form reflects not the object as understood phenomenologically, but the object as a relational system of measurable differences across viewpoints. In this sense, the reconstruction makes visible a fundamentally structuralist logic: the identity of the lemon does not arise from an essential or self-contained objecthood, but from a network of relations, between light conditions, images features, spatial correspondences, and perceptual inferences, that together produce recognition. What appears as a coherent object is therefore not simply given, but constructed through structures. At the same time, the distortions and flattened form of the reconstructed lemon show the limits of relying only on structural relationships, suggesting that representations built through algorithmic correlations can remain disconnected from lived, sensory experience. In this sense, machine vision reflects the broader logic of generative AI, where reality is not directly captured but instead reconstructed through statistical patterns learned from previous images. The lemon becomes not an object but a diagram of relations, demonstrating how both optical media and computational systems translate the world into abstract structures governed by geometry, probability, and transformation.

To further examine how machines represent images, I extended this investigation into the internal processes of generative AI by creating a visualization of image formation within a Stable Diffusion model. Stable Diffusion belongs to a class of generative models known as diffusion models, which produce images through a gradual transformation from noise into recognizable structure.

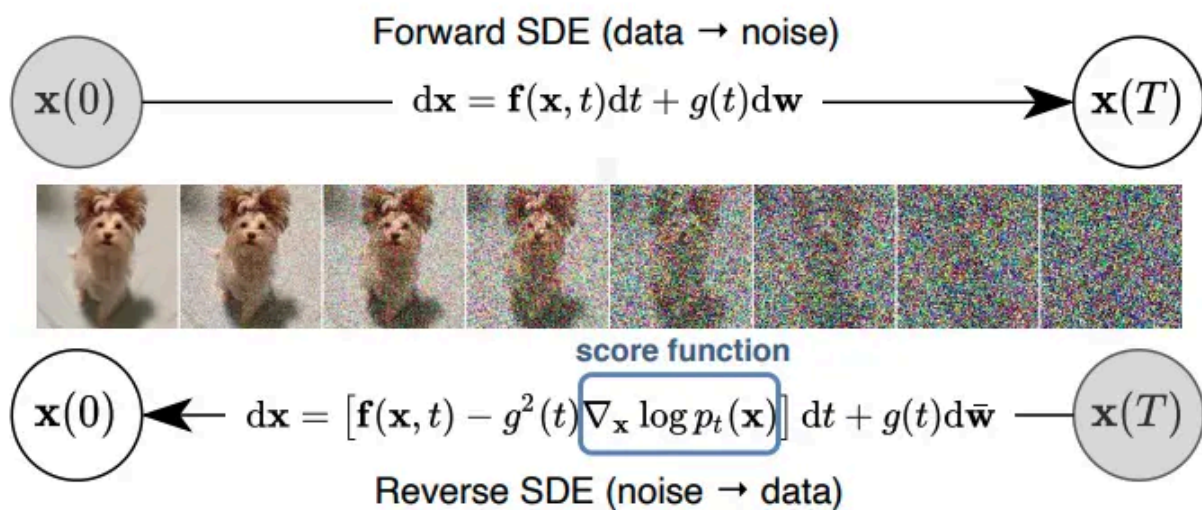


Figure 3. The mathematical process and visualization of a stable diffusion model

Instead of constructing an image directly from an explicit representation of the world, the model begins with a field of random noise and iteratively refines it through a sequence of probabilistic steps (fig. 3). During training, the model learns how images become progressively noisier as random perturbations are added. To generate a new image, the process is reversed: the model starts with noise and slowly removes randomness by predicting the structural patterns most likely to correspond to the given textual prompt. Each step slightly reorganizes the distribution of pixels according to learned statistical relationships, allowing visual elements such as edges, textures, and spatial arrangements to emerge gradually from an initially chaotic field.

What is significant in this process is that the intermediate states do not resemble any recognizable object from a human perspective. The latent representations within the diffusion

process are highly abstract, composed of multidimensional vectors that encode probabilistic relationships between visual features rather than directly corresponding to objects in the world. These intermediate forms exist within what is often referred to as *latent space*, a mathematical structure in which concepts are organized through similarity and difference. In latent space, categories such as “apple,” “chair,” or “street” are not stored as discrete visual templates but as clusters of statistical relationships between features learned across millions of training images. The model does not know what an apple is in the experiential sense; it only encodes correlations between color distributions, shapes, textures, and contextual associations that frequently co-occur within the dataset.

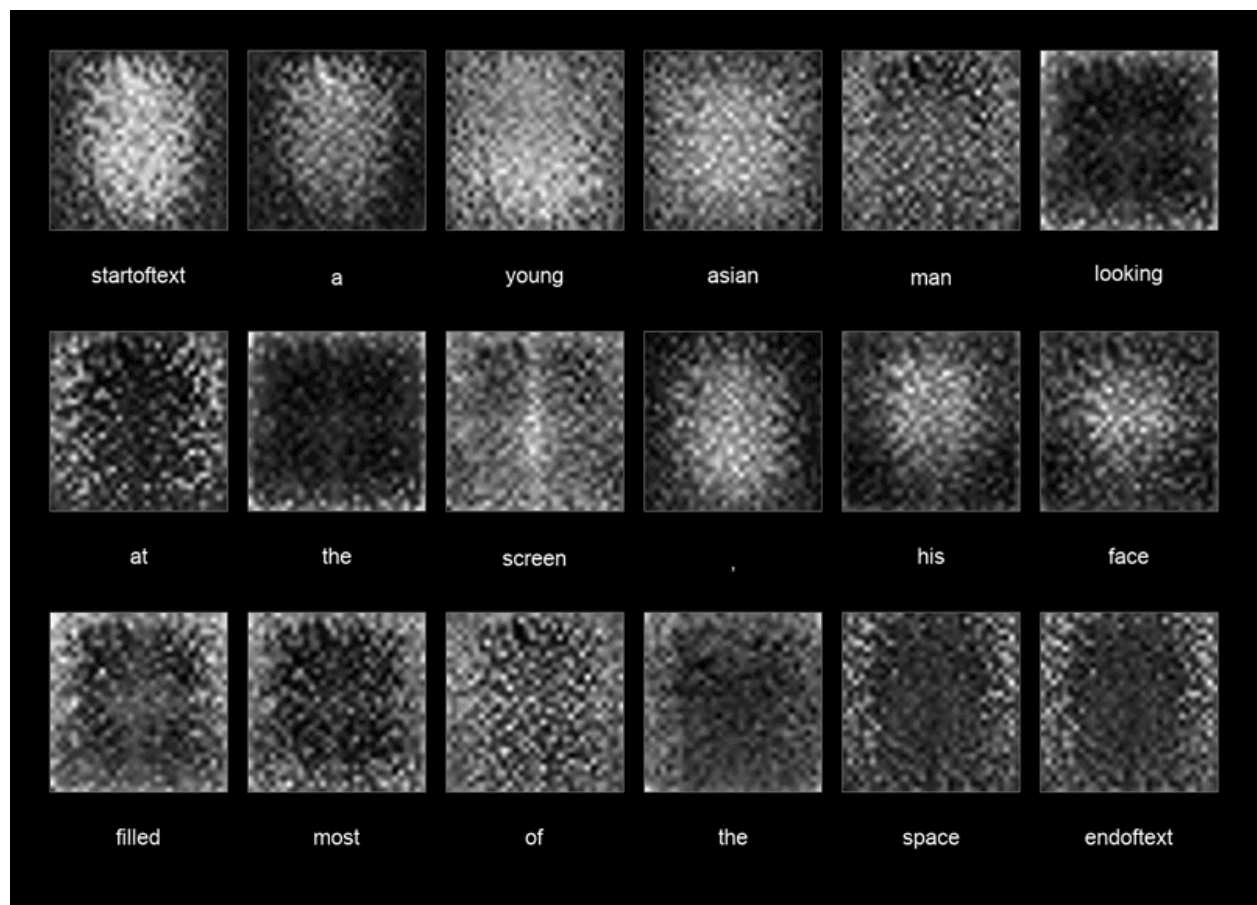


Figure 4. The latent space of the prompt “a young Asian man looking at the screen, his face filled most of the space”

This form of representation can be understood as a distinct mode of abstraction, one operating less through visible form than through encoded relations. Rather than opposing machine abstraction to artistic abstraction, this work considers how abstraction can function through different representational logics. In many artistic practices, abstraction has served to reduce, distort, systematize, or reconfigure perception in order to reveal structures not immediately visible in the world. Latent representations in generative AI can be understood as extending this question into a computational domain, where abstraction occurs through numerical compression and probabilistic encoding. The latent representation is not meant to be seen directly, yet it contains the structural information necessary to reconstruct an image. In this sense, latent space can be understood as a domain of representation that is legible primarily to machines. Like Trevor Paglen in *Invisible Image* argues that machine vision has its own visual regime and their images are illegible to humans (Paglen 2016), the latent space is a form of coded visual culture shared among neural networks, composed not of symbols or images but of weighted relationships between dimensions. Within this encoded environment, visual coherence is distributed across parameters and emerges only when translated back into visual form through decoding processes.

The visualization of latent noise transition in this project attempts to make perceptible this otherwise invisible representational domain. By projecting the gradual emergence of an image from noise, the work stages a temporal encounter between human perception and machine perception. What appears initially as meaningless noise slowly reveals coherent structure, demonstrating how recognizable forms can arise from statistical relationships rather than direct observation. For the human viewer, this transition may appear as a movement from abstraction toward representation; for the machine, however, the process operates continuously within an abstract representational framework. The generated image is not derived from a stored picture but synthesized through relational probabilities embedded in the model's learned parameters.

Within the broader context of the project, this latent transition demonstrates how a recognizable image can dissolve into its underlying statistical structure without fully losing its semantic coherence. Just as the projection mapping of the painted scene reveals the persistence of structural cues beneath abstraction, the diffusion visualization reveals that AI image generation depends on an underlying structural organization that does not resemble the final image. The latent state should not be understood as a residual trace of the image, but as its underlying structural form, demonstrating that what appears as visual coherence is generated from layers of abstract relational encoding. This suggests that both human and machine perception rely on structured forms of inference, yet the structures themselves may operate at different levels of visibility. By presenting the diffusion process as an aesthetic and perceptual experience, the project highlights how contemporary image culture increasingly depends on representations produced through systems that do not perceive the world directly but construct it through abstract relational patterns.

Reversely, I deployed an object detection model to investigate a complementary condition: forms of abstraction that remain legible to humans yet become increasingly difficult for AI systems to interpret. Object detection models such as YOLO or Faster R-CNN operate by scanning an image and identifying visual features that statistically correspond to categories learned during training. The model processes the image through a convolutional neural network, extracting hierarchical feature maps that capture edges, textures, and spatial patterns. These features are compared to learned representations associated with predefined classes such as “person,” “face,” or “chair.” The model then assigns a confidence score indicating the probability that a region of the image belongs to a specific category. Importantly, these systems rely heavily on consistency between training data and input images; when visual structure deviates too far from learned statistical patterns, the model’s confidence decreases or fails entirely.

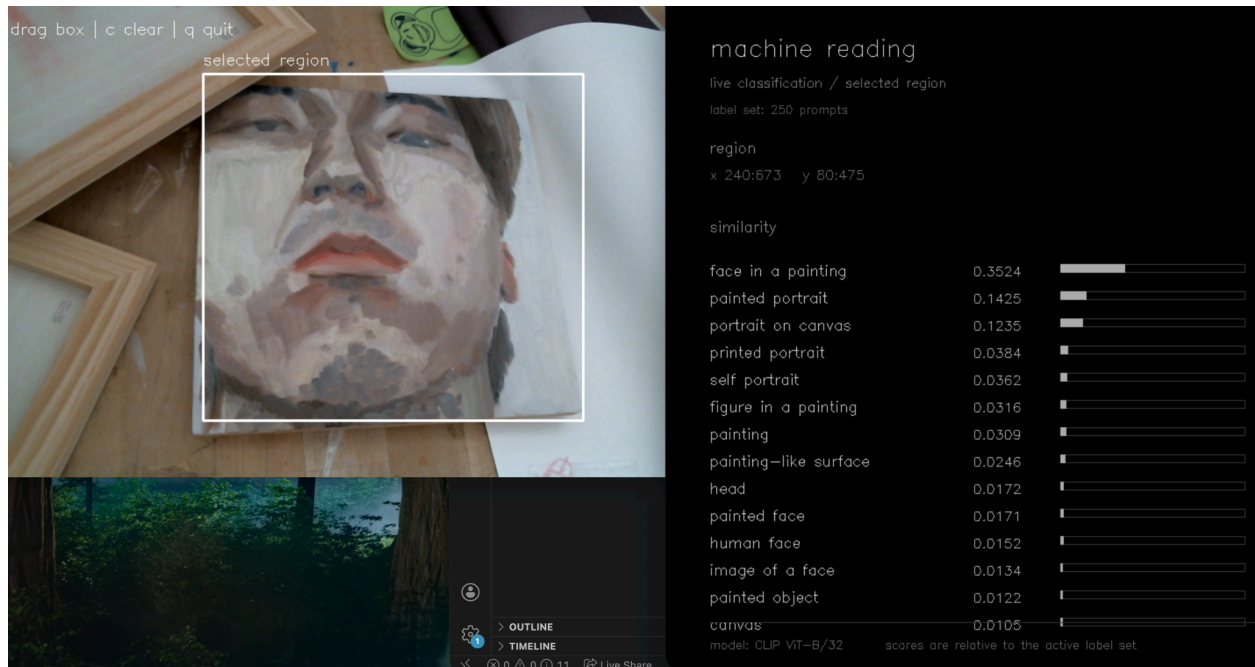


Figure 5. The visual interface of object detection software and the self-portrait

In this experiment, I created an abstract painting based on a portrait of my own face and projected a realistic photographic image onto the painting. Over time, the projected image gradually dissolves into abstraction until it visually merges with the painted surface. Beneath the image, a real-time confidence score from the object detection model displays the probability that the system recognizes the subject as a face or person. As the projection becomes increasingly abstract, the confidence score steadily drops, demonstrating the limits of machine recognition when familiar statistical features are no longer clearly present. However, human viewers are still able to recognize the portrait even when the image loses much of its descriptive fidelity. This persistence of recognition reveals an important distinction between human perception and machine perception. While both humans and AI rely on prior information, human perception draws on embodied experience, contextual awareness, and cultural knowledge—the accumulated warmth of lived experience—whereas AI systems operate through statistical associations derived from training data. Even when facial details dissolve into color fields or

fragmented geometry, viewers can rely on prior familiarity with facial structure and the contextual expectation of portraiture to maintain recognition (Gombrich 1960).

Because the painting is derived from my own selfie, an additional layer of meaning emerges through social relations. For viewers who know me personally, recognition may persist even when the image becomes highly abstract, supported by memories of my appearance and shared interpersonal context. In this sense, meaning is reinforced through networks of personal and cultural associations rather than purely visual information. Recognition becomes distributed across social knowledge, memory, and expectation, demonstrating that perception operates within a broader relational system extending beyond the image itself.

Compared to the latent noise transition experiment, which visualizes a form of abstraction primarily legible to machines, this work presents an inverse situation in which abstraction remains meaningful for humans but becomes increasingly illegible to AI. In the diffusion process, intermediate latent states appear as noise to human viewers, yet they contain structured information that the model can interpret and transform into coherent images. The abstraction exists within a mathematical representational space shared among machine learning systems. In contrast, the abstract portrait retains recognizable structure for human viewers because it resonates with cognitive schemas of facial organization and with cultural familiarity with portraiture. The AI model, however, struggles to interpret this abstraction because it lacks access to these broader contextual networks and instead depends on statistical consistency with its training distribution.

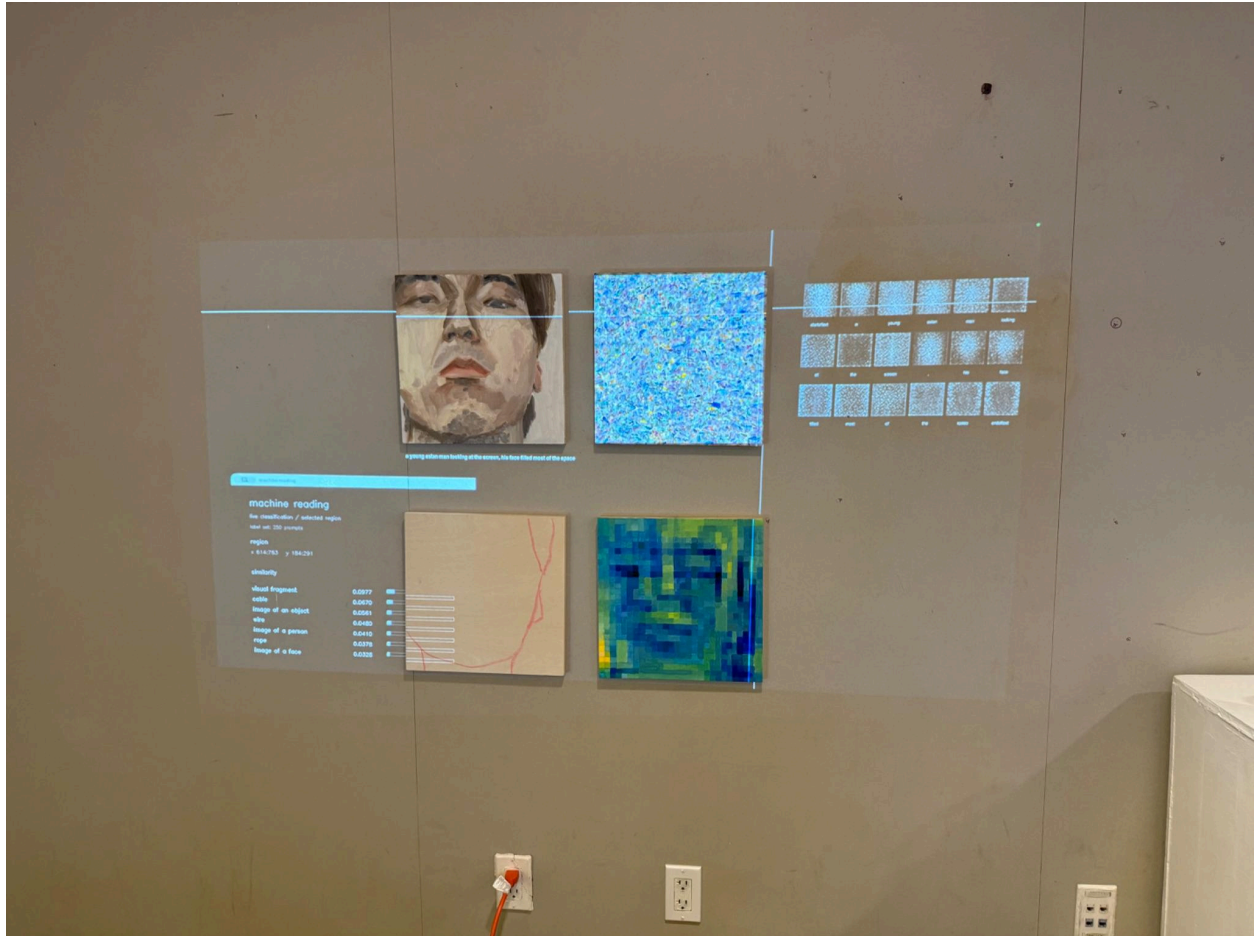


Figure 6. The final product

Together, these two experiments reveal a divergence between human and machine abstraction: one demonstrates how machines operate within an abstract representational domain that is not intuitively accessible to humans, while the other demonstrates how human perception can sustain meaning even when visual correspondence becomes minimal. Through this comparison, abstraction becomes a site where different perceptual systems—biological and computational—reveal their respective structures, limitations, and modes of understanding.

Despite the strong theoretical foundation and extensive research behind the project, the physical presentation of the work is still not fully resolved. While the concepts surrounding structuralism, perception, and AI representation are clearly developed throughout the research and writing, the installation itself does not always communicate these ideas as clearly as

intended. Some parts of the projection mapping and object detection visualization remain visually subtle, making the work occasionally difficult to understand without additional explanation. This creates a gap between the conceptual depth of the project and its material execution. The limited scale of the painting, the projection quality, and the installation setup also reduce the immersive impact the work aims to create. In future developments, the project could benefit from a larger painting, more precise projection mapping, and a more refined installation environment that allows the transitions between realism, abstraction, and machine perception to become more visually immediate and experiential. At the same time, these limitations also reflect one of the central concerns of the project itself: the difficulty of translating invisible computational structures and abstract philosophical ideas into physical and perceptible forms. In this sense, the occasional obscurity of the installation is not only a technical limitation, but also part of the challenge of making machine perception and relational structures visible to human viewers.

Work Cited:

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